

Balaji Ramachandran

Mathematics PhD Student · Carnegie Mellon University

[Academic website](#) [Google Scholar](#) [LinkedIn](#)

RESEARCH INTERESTS

Stochastic analysis and partial differential equations; generative diffusion modelling; high-dimensional and multimodal sampling; reinforcement learning; mathematical finance; stochastic games, control, and optimization.

EDUCATION

Carnegie Mellon University

June 2026 – present

PhD student, Mathematics

Research direction: mathematical finance.

Indian Institute of Science

From 2022

BTech, Mathematics and Computer Science

Bengaluru, India

PUBLICATIONS

A Network Formation Game for Katz Centrality Maximization: A Resource Allocation Perspective

Balaji R., Prashil Wankhede, Pavankumar Tallapragada.

Accepted at the 65th IEEE Conference on Decision and Control (CDC), 2026. Honolulu, Hawaii, December 15–18.

Presenting in person.

arXiv:2604.03056 · doi:10.48550/arXiv.2604.03056

RESEARCH AND PROFESSIONAL EXPERIENCE

HSBC

2026

Quantitative analyst intern

Worked on stress-scenario risk management dashboards and portfolio risk analysis for the FRTB team.

Purdue University

June – August 2025

Graduate research intern

Developed a physics-informed neural network solution for a PDE to estimate diffusion hitting times. Worked on nonparametric impulse control of diffusion processes.

Indian Institute of Science

From April 2024

Student researcher

Also served as student coordinator from May 2024.

DISTINCTIONS

JEE Advanced · All India Rank 225.

